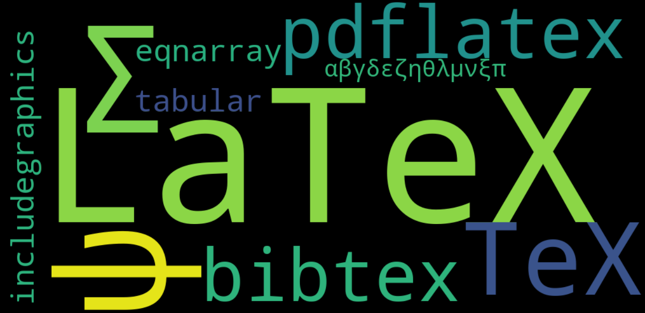
GETTING STARTED WITH $\LaTeX$



What is LaTeX?

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Author: Leslie Lamport, available to public since 1984

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FREE!

How to install LaTeX?

Windows	Go to https://miktex.org/download and download the basic installer. For detailed instructions, refer to https://miktex.org/howto/install-miktex.
Linux	Install *texlive* and use your favourite editor.
android	Install VerbTeX.

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Or, go to https://www.overleaf.com.

The simplest TEX file

```
\documentclass{article}
\begin{document}
Let's learn \LaTeX.
\end{document}
```

The simplest TEX file

Input

```
\documentclass{article}
\begin{document}
Let's learn \LaTeX.
\end{document}
```

Output

Let's learn LATEX.

Compiled using *pdflatex*.

Essential parts of a TEX file

Preamble	`\documentclass[optional arguments]{article/letter/book/...}`

Body	`\begin{document}`
	Body of the document
	`\end{document}`

Adding some basic stuff

```
\documentclass[a4paper,12pt]{article}
\title{My Laws}
\author{Isaac Newton}
\date{January 1, 1687}
\begin{document}
\maketitle
\section{First Law}
A body at rest will continue to be at rest and
a body in uniform motion will stay in uniform
motion unless acted upon by an external force.
\section{Second Law}
The rate of change of momentum is directly
proportional to the impressed force.
\section{Third Law}
Every action has an equal and opposite
reaction.
\end{document}
```

My Laws

Isaac Newton

January 1, 1687

1 First Law

A body at rest will continue to be at rest and a body in uniform motion will stay in uniform motion unless acted upon by an external force.

2 Second Law

The rate of change of momentum is directly proportional to the impressed force.

3 Third Law

Every action has an equal and opposite reaction.

Subsections, text formatting etc

```
\documentclass{article}
\begin{document}
\section{Optics}
This section is about {\bf optics}.
\subsection{Geometrical Optics}
This is the {\em small wavelength limit}.

\subsection{Physical Optics}
Here, we will discuss \underline{physical optics}.
\subsubsection{Diffraction}
\paragraph{Fraunhofer Diffraction}
Sources and screens are \textsc{far away}.
\paragraph{Fresnel Diffraction}
Sources and/or screens are nearby.
\end{document}
```

Subsections, text formatting etc

Input

```
\documentclass{article}
\begin{document}
\section{Optics}
This section is about {\bf optics}.
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Sources and screens are \textsc{far away}.
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\end{document}
```

Output

1 Optics

This section is about **optics**.

1.1 Geometrical Optics

This is the *small wavelength limit*.

1.2 Physical Optics

Here, we will discuss physical optics.

1.2.1 Diffraction

Fraunhofer Diffraction Sources and screens are FAR AWAY.

Fresnel Diffraction Sources and/or screens are nearby.

Font sizes

```
0 {\tiny 1 \scriptsize 2 \footnotesize 3 \small 4 \normalsize 5
  \large 6 \Large 7 \LARGE 8 \huge 9 \Huge 10} 11
```

Font sizes

```
0 {\tiny 1 \scriptsize 2 \footnotesize 3 \small 4 \normalsize 5
  \large 6 \Large 7 \LARGE 8 \huge 9 \Huge 10} 11
```

0 1 2 3 4 5 6 7 8 9 10 11

New commands/Macros

```
\newcommand{\myname}{Saurish Chakrabarty}
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I am \myname.
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This can be put in the preamble.

```
I am \myname.
```

I am Saurish Chakrabarty.

```
\newcommand{\lastnamefirst}[2]{\textsc{#2}, {\em #1}}
```

New commands/Macros

```
\newcommand{\myname}{Saurish Chakrabarty}
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This can be put in the preamble.

```
I am \myname.
```

I am Saurish Chakrabarty.

```
\newcommand{\lastnamefirst}[2]{\textsc{#2}, {\em #1}}
```

```
\lastnamefirst{Satyendra Nath}{Bose}\\
\lastnamefirst{Albert}{Einstein}
```

BOSE, *Satyendra Nath*
EINSTEIN, *Albert*

Some more things

- Comments: `%This is a comment. It will not appear in the pdf file.`

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- Using reserved symbols: %, &, #, {, }, etc can be printed using the backslash character. *E.g.*, 100\% prints 100%.

Mathematical Expressions

Basic math

Inline:

```
\(x=\pi^2\)
```

Basic math

Inline:

`\(x=\pi^2\)` becomes $x = \pi^2$

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The same can be done using `$x=\pi^2$`.

Displayed equation/formula:

```
\[
  n_{\vec{k}}=
  \frac{1}{e^{\beta(E_{\vec{k}}-\mu)}\pm 1}
\]
```

$$n_{\vec{k}}=\frac{1}{e^{\beta(E_{\vec{k}}-\mu)}\pm 1}$$

Inline:

`\(x=\pi^2\)` becomes $x = \pi^2$

The same can be done using `$x=\pi^2$`.

Displayed equation/formula:

```
\[
  n_{\vec{k}}=
  \frac{1}{e^{\beta(E_{\vec{k}}-\mu)}\pm 1}
\]
```

$$n_{\vec{k}} = \frac{1}{e^{\beta(E_{\vec{k}}-\mu)} \pm 1}$$

```
\begin{equation}
  -\frac{\hbar^2}{2m}\psi''+V\psi=E\psi
\end{equation}
```

$$-\frac{\hbar^2}{2m}\psi'' + V\psi = E\psi \qquad (1)$$

Calligraphic fonts, ellipsis, and some more symbols in math mode

\mathcal{A,B,C,\ldots,X,Y,Z}

$\mathcal{A,B,C,\ldots,X,Y,Z}$

Calligraphic fonts, ellipsis, and some more symbols in math mode

\mathcal{A,B,C,\ldots,X,Y,Z} $\mathcal{A,B,C,\ldots,X,Y,Z}$

\sum_{n=1}^Nn=\frac{N(N+1)}{2}

$$\sum_{n=1}^N n=\frac{N(N+1)}{2}$$

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$$\sum_{n=1}^Nn=\frac{N(N+1)}{2}$$

\frac{1}{\sigma\sqrt{2\pi}}
\int_{-\infty}^\infty\exp\left[-\frac{1}{2}
\left(\frac{x-\mu}{\sigma}\right)^2\right]

$$\frac{1}{\sigma\sqrt{2\pi}}\int_{-\infty}^\infty\exp\left[-\frac{1}{2}\left(\frac{x-\mu}{\sigma}\right)^2\right]$$

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```

$$\frac{1}{\sigma\sqrt{2\pi}}\int\limits_{-\infty}^\infty\exp\left[-\frac{1}{2}\left(\frac{x-\mu}{\sigma}\right)^2\right]$$

Multiple equations

```
\begin{eqnarray}
A x = B \\
\Rightarrow x = A^{-1}B
\end{eqnarray}
```

$$Ax = B \tag{2}$$

$$\Rightarrow x = A^{-1}B \tag{3}$$

Multiple equations

```
\begin{eqnarray}
A x = B \\
\Rightarrow x = A^{-1}B
\end{eqnarray}
```

$$
\begin{aligned}
Ax = B \qquad (2)\\
\Rightarrow x = A^{-1}B \qquad (3)
\end{aligned}
$$

```
\begin{eqnarray}
\mu &=& \langle x\rangle \\
\Rightarrow \sigma^2&=&
\left\langle x^2\right\rangle-\mu^2
\label{variance}
\end{eqnarray}
```

$$
\begin{aligned}
\mu &= \langle x\rangle \qquad (4)\\
\Rightarrow \sigma^2 &= \left\langle x^2\right\rangle-\mu^2 \qquad (5)
\end{aligned}
$$

Multiple equations

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\begin{eqnarray}
A x = B \\
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\end{eqnarray}
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$$
\begin{aligned}
Ax &= B && (2)\\
\Rightarrow x &= A^{-1}B && (3)
\end{aligned}
$$

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\end{eqnarray}
```

$$
\begin{aligned}
\mu &= \langle x\rangle && (4)\\
\Rightarrow \sigma^2 &= \left\langle x^2\right\rangle-\mu^2 && (5)
\end{aligned}
$$

\nonumber suppresses the equation number.

Arrays in math mode

```
A=\left(
\begin{array}{ccc}
11 & 12 & 13\\
21 & 22 & 23\\
31 & 32 & 33\\
\end{array}
\right)
```

$$A = \left(\begin{array}{ccc} 11 & 12 & 13 \\ 21 & 22 & 23 \\ 31 & 32 & 33 \\ \end{array} \right)$$

Arrays in math mode

```
A=\left(
\begin{array}{ccc}
11 & 12 & 13\\
21 & 22 & 23\\
31 & 32 & 33\\
\end{array}
\right)
```

$$A = \left(\begin{array}{ccc} 11 & 12 & 13 \\ 21 & 22 & 23 \\ 31 & 32 & 33 \end{array} \right)$$

```
\Theta(x)=\left\{
\begin{array}{ll}
0, & \mbox{if }x<0,\mbox{ and,}\\
1, & \mbox{if }x\ge0\\
\end{array} \right.
```

$$\Theta(x) = \left\{ \begin{array}{ll} 0, & \mbox{if } x < 0, \mbox{ and,} \\ 1, & \mbox{if } x \ge 0 \end{array} \right.$$

Use of AMSmath package

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- Check out *multiline*, *gather*, *align*, and, *alignat*.
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```
\begin{matrix}
1&0\\
0&1\\
\end{matrix}
```

$$\begin{matrix} 1 & 0 \\ 0 & 1 \end{matrix}$$

Use of AMSmath package

- \usepackage{amsmath}
- Check out *multiline*, *gather*, *align*, and, *alignat*.
- For matrices, amsmath has *matrix*, *pmatrix*, *bmatrix*, *vmatrix*, *Vmatrix*, and, *smallmatrix*.

```
\begin{pmatrix}
1&0\\
0&1\\
\end{pmatrix}
```

$$\begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}$$

Use of AMSmath package

- \usepackage{amsmath}
- Check out *multiline*, *gather*, *align*, and, *alignat*.
- For matrices, amsmath has *matrix*, *pmatrix*, *bmatrix*, *vmatrix*, *Vmatrix*, and, *smallmatrix*.

```
\begin{vmatrix}
1&0\\
0&1\\
\end{vmatrix}
```

$$\begin{vmatrix} 1 & 0 \\ 0 & 1 \end{vmatrix}$$

Referring to equations

Equations can be labeled using the \label{} command and can be referred to using the \ref{} command.

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Equation \ref{variance} can be used to calculate the variance.

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Equation \ref{variance} can be used to calculate the variance.

$\longrightarrow$ Equation 5 can be used to calculate the variance.

Tables

```
\begin{tabbing}
  \hspace{40pt} \= Chess \= Ludo \= Carrom \= Bridge \\
  Pieces \> 32 \> 17 \> 20 \> 52\\
  Players \> 2 \> 4 \> 4 \> 4\\
\end{tabbing}
```

	Chess	Ludo	Carrom	Bridge
Pieces	32	17	20	52
Players	2	4	4	4

Tables

```
\begin{tabular}{|c|c|c|c|}
  \hline
  Particle & electron & proton & neutron\\
  \hline
  Mass & 0.5 & 935 & 935\\
  Charge & -1 & +1 & 0\\
  \hline
\end{tabular}
```

Tables

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\begin{tabular}{|c|c|c|c|}
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Particle	electron	proton	neutron
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Particle	electron	proton	neutron
Mass	0.5	935	935
Charge	-1	+1	0

Column options

l: left aligned, *r*: right aligned, *c*: centered,
p{⟨*width*⟩}: justified to specified width.

Tables (adding a caption and a label)

```latex
\begin{table}
  \begin{tabular}{|c|c|c|c|}
    \hline
    Particle & electron & proton &
                         neutron\\
    \hline
    Mass & 0.5 & 935 & 935\\
    Charge & -1 & +1 & 0\\
    \hline
  \end{tabular}
  \caption{Properties of elementary
    particles}
  \label{chargeMass}
\end{table}
Table \ref{chargeMass} contains some
properties of some elementary particles.
```

Tables (adding a caption and a label)

```
\begin{table}
  \begin{tabular}{|c|c|c|c|}
    \hline
    Particle & electron & proton &
                          neutron\\
    \hline
    Mass & 0.5 & 935 & 935\\
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  \end{tabular}
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OUTPUT

Particle	electron	proton	neutron
Mass	0.5	935	935
Charge	-1	+1	0

Table 1: Properties of elementary particles

Table 1 contains some properties of some elementary particles.

Pictures

\usepackage{graphicx}

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\usepackage{graphicx}
\includegraphics[width=⟨*width*⟩]{⟨*filename*⟩}

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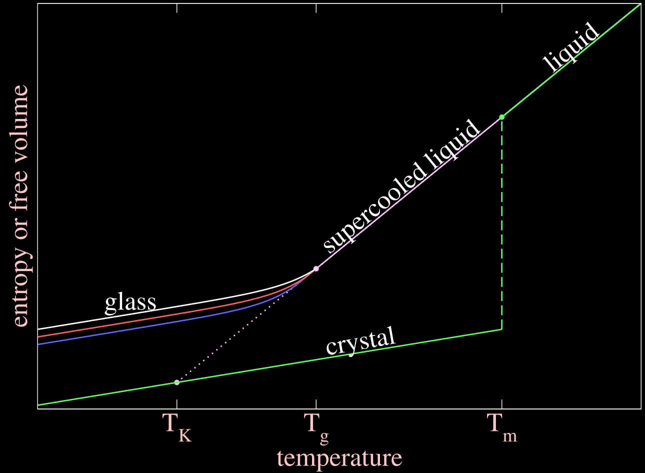
\usepackage{graphicx}

\includegraphics[width=⟨*width*⟩]{⟨*filename*⟩}



Pictures (adding a caption and a label)

\begin{figure}
⟨*includegraphics*⟩
\caption{⟨*caption*⟩}
\label{landscapes}
\end{figure}

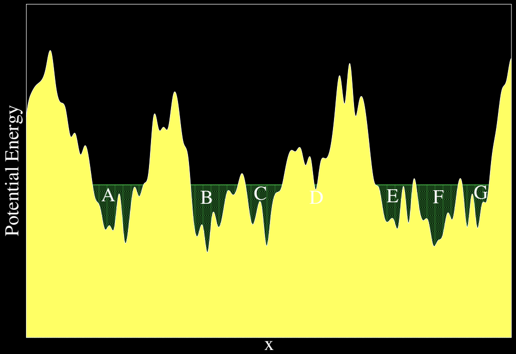


Figure 1: Energy landscapes of supercooled liquids

Pictures (adding a caption and a label)

\begin{figure}
⟨*includegraphics*⟩
\caption{⟨*caption*⟩}
\label{landscapes}
\end{figure}

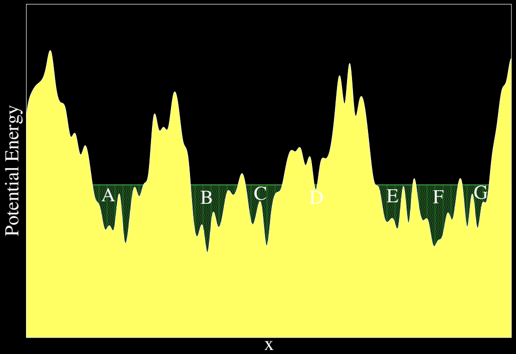


Figure 1: Energy landscapes of supercooled liquids

The energy landscape is drawn in Fig. \ref{landscapes}.

\begin{figure}
⟨*includegraphics*⟩
\caption{⟨*caption*⟩}
\label{landscapes}
\end{figure}

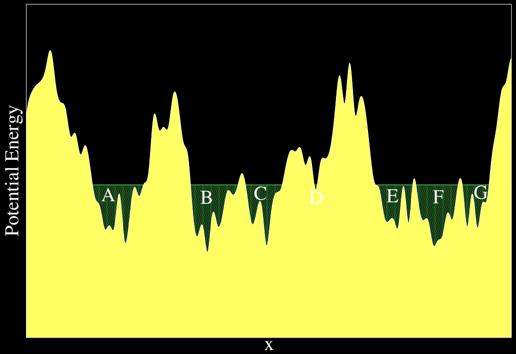


Figure 1: Energy landscapes of supercooled liquids

The energy landscape is drawn in Fig. \ref{landscapes}.
$\longrightarrow$ The energy landscape is drawn in Fig. 1.

Lists

```latex
\begin{itemize}
\item Classical Mechanics
\item Thermodynamics
\item Statistical Mechanics
\item Electrodynamics
\item Quantum Mechanics
\end{itemize}
```

- Classical Mechanics
- Thermodynamics
- Statistical Mechanics
- Electrodynamics
- Quantum Mechanics

Lists (nested)

```latex
\begin{itemize}
\item Classical Mechanics
\item Thermodynamics
\item Statistical Mechanics
  \begin{itemize}
  \item Classical Statistical Mechanics
  \item Quantum Statistical Mechanics
    \begin{itemize}
    \item Bose-Einstein statistics
    \item Fermi-Dirac statistics
    \end{itemize}
  \end{itemize}
\item Electrodynamics
\item Quantum Mechanics
\end{itemize}
```

- Classical Mechanics
- Thermodynamics
- Statistical Mechanics
  - Classical Statistical Mechanics
  - Quantum Statistical Mechanics
    - Bose-Einstein statistics
    - Fermi-Dirac statistics
- Electrodynamics
- Quantum Mechanics

Lists (numbered)

```
\begin{enumerate}
\item Classical Mechanics
\item Thermodynamics
\item Statistical Mechanics
  \begin{enumerate}
  \item Classical Statistical Mechanics
  \item Quantum Statistical Mechanics
    \begin{enumerate}
    \item Bose-Einstein statistics
    \item Fermi-Dirac statistics
    \end{enumerate}
\   end{enumerate}
\item Electrodynamics
\item Quantum Mechanics
\end{enumerate}
```

1. Classical Mechanics
2. Thermodynamics
3. Statistical Mechanics
    a. Classical Statistical Mechanics
    b. Quantum Statistical Mechanics
        i. Bose-Einstein statistics
        ii. Fermi-Dirac statistics
4. Electrodynamics
5. Quantum Mechanics

Bibliography using BibTeX

```
\documentclass{article}
\begin{document}
In vacuum, nothing can travel faster
than light.\cite{einsteinSTR} In some
medium, Cherenkov observed emission of
radiation when a charged particle
moved faster than the speed of light
in that medium.\cite{cherenkov}
\bibliography{sample}
\bibliographystyle{plain}
\end{document}
```

Bibliography using BibTeX

```latex
\documentclass{article}
\begin{document}
In vacuum, nothing can travel faster
than light.\cite{einsteinSTR} In some
medium, Cherenkov observed emission of
radiation when a charged particle
moved faster than the speed of light
in that medium.\cite{cherenkov}
\bibliography{sample}
\bibliographystyle{plain}
\end{document}
```

sample.bib

```bibtex
@article{einsteinSTR,
  title={On the electrodynamics of moving bodies},
  author={Einstein, Albert},
  journal={Annalen der physik},
  volume={17},
  number={10},
  pages={891--921},
  year={1905}
}

@article{cherenkov,
  title={Visible light from clear liquids under
  the action of gamma radiation},
  author={Cherenkov, Pavel A},
  journal={Comptes Rendus (Doklady) de l'Aeademie
  des Sciences de l'URSS},
  volume={2},
  number={8},
  pages={451--454},
  year={1934}
}
```

Bibliography using BibTeX

Compilation order *pdfLaTeX* $>$ *bibTeX* $>$ *pdfLaTeX* $>$ *pdfLaTeX*
(Ignore if you are using overleaf)

Compilation order *pdfLaTeX* > *bibTeX* > *pdfLaTeX* > *pdfLaTeX*
(Ignore if you are using overleaf)

In vacuum, nothing can travel faster than light.[2] In some medium, Cherenkov observed emission of radiation when a charged particle moved faster than the speed of light in that medium.[1]

References

[1] Pavel A Cherenkov. Visible light from clear liquids under the action of gamma radiation. *Comptes Rendus (Doklady) de l'Aeademie des Sciences de l'URSS*, 2(8):451–454, 1934.

[2] Albert Einstein. On the electrodynamics of moving bodies. *Annalen der physik*, 17(10):891–921, 1905.

Miscellaneous

- Colors
  - `\usepackage{color}`
  - `\textcolor{red}{Apple}` $\rightarrow$ Apple
  - `\definecolor{darkRed}{RGB}{150,0,0}`

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  - `\usepackage{color}`
  - `\textcolor{red}{Apple}` $\rightarrow$ Apple
  - `\definecolor{darkRed}{RGB}{150,0,0}`
- Table of contents
  - Used in documentclass *book*
  - `\tableofcontents`

Miscellaneous

- Page size and margins

Miscellaneous

- Page size and margins
  - `\usepackage{geometry}`
  - ```
    \geometry{paperheight=14cm,paperwidth=20cm}
    \geometry{tmargin=-0.3in,bmargin=0.3in}
    \geometry{lmargin=0.5in,rmargin=0.5in}
    ```
    ```

Miscellaneous

- Page size and margins
  - `\usepackage{geometry}`
  - `\geometry{paperheight=14cm,paperwidth=20cm}`
    `\geometry{tmargin=-0.3in,bmargin=0.3in}`
    `\geometry{lmargin=0.5in,rmargin=0.5in}`
  - `\geometry{a4paper,total={17cm,25.7cm},top=0.5cm,left=2cm}`

Miscellaneous

- Page size and margins
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    `\geometry{lmargin=0.5in,rmargin=0.5in}`
  - `\geometry{a4paper,total={17cm,25.7cm},top=0.5cm,left=2cm}`
- Stretching/Shrinking: `\renewcommand{\baselinestretch }{0.9}`

Miscellaneous

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    `\geometry{tmargin=-0.3in,bmargin=0.3in}`
    `\geometry{lmargin=0.5in,rmargin=0.5in}`
  - `\geometry{a4paper,total={17cm,25.7cm},top=0.5cm,left=2cm}`
- Stretching/Shrinking: `\renewcommand{\baselinestretch }{0.9}`
- arXiv: Download TEX files of articles written by your favorite researcher. (They are in *tar* format. Install suitable software to untar.)

References

- Stefan Kottwitz, *LaTeX Beginner's Guide*
- Scott Pakin, *The Comprehensive LaTeX Symbol List*
- Overleaf – https://www.overleaf.com
- arXiv – https://arxiv.org

☺ Thank You ☺

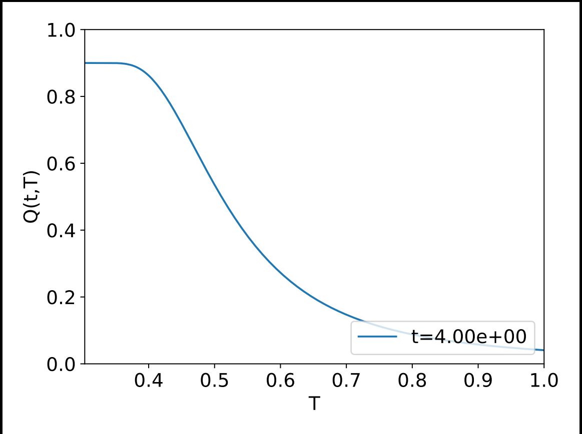
☺ Thank You ☺

Homework ☺

Create class notes of your favorite subject. Try to put at least one equation, at least one figure, at least one table and a bibliography. Share the notes with your teacher and incorporate whatever feedback you get.

Further questions and discussions: saurish@apccollege.ac.in

Video Embedding



Template containing video embedding: https://www.overleaf.com/read/rcfqksxktygc

Using tikz

Computational Physics

ODE based models | PDE based models | SDE based models